

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours

FULL MARKS: 210

- N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

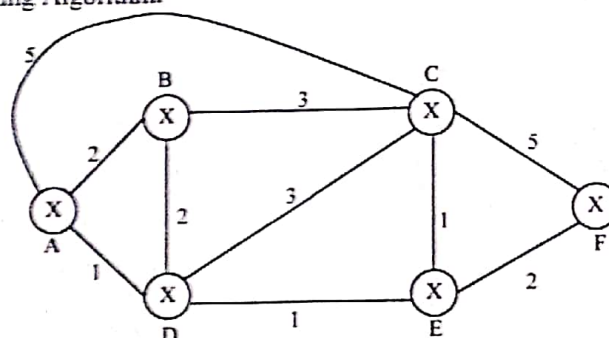
SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Briefly discuss about client-server architecture. Discuss about the major challenges of P2P architecture. (12)
- b) "Socket is analogous to door" – Justify the statement. What transport service does an app need? (08)
- c) What are the components of cookies? Describe how cookies keep state using database with proper examples. (10)
- d) FTP control connection is "out of band". Explain. (05)
2. a) Let, host at *abc.poly.edu* wants IP address for *defg.cs.xyz.edu*. Show, how it is done using recursive queries for various DNS servers. (09)
- b) Explain the necessity of mail access agent in email system. Compare and contrast the functionalities of POP and IMAP. (10)
- c) Consider the following 20-byte header (IPv4). Find the header checksum. Why does IPv4 header checksum cover only data portion? (10)

4	5	0	28	
1			0	0
4	17		0	
10.12.14.5				
12.6.7.9				

- d) How does conditional GET command improve performance of web proxy and local cache mechanism? (06)
3. a) Explain how DHCP works. What are the main benefits of DHCP? (09)
- b) Apply fragmentation on following information for IPv4 datagram. (13)
 - i) Original Datagram: 5000 bytes, Header: 20 bytes.
 - ii) MTU: 1st level 2000 bytes & 2nd level: 800 bytes.
- c) An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to 3 groups of customers: (13)
 - i) 1st group: 52 customers, each needs 128 addresses.
 - ii) 2nd group: 40 customers, each needs 256 addresses.
 - iii) 3rd group: 100 customers, each needs 64 addresses. How many addresses are still unallocated? Note down the network addresses of each group.
4. a) Briefly explain how PAT works. Why is PAT used over NAT? (10)
- b) Determine the routing table/forwarding table at the router B from the following topology using a Link-state Routing Algorithm. (17)



- c) Concisely describe the error reporting messages of ICMP. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Describe the working principle of Ethernet's CSMA/CD protocol with a detail look on exponential back-off algorithm. (10)
b) Write filtering/forwarding algorithm of a switch. List the actions with corresponding conditions that a switch may perform when it receives a frame. (09)
c) What is multiprotocol label switching (MPLS)? How does it work? Describe MPLS functionality with a simple example. (12)
d) Describe silly windows syndrome created by the sender. (04)
6. a) How does TCP handle congestion? Describe with proper state diagram and graph. (15)
b) Write down the connection establishment and half close in TCP. (10)
c) How retransmission timeout is calculated in TCP? In which situation Karn's algorithm is applied? (10)
7. a) Briefly explain how Secure Socket Layer (SSL) works on top of TCP and what services it provides. (10)
b) What is the difference between RDT 2.2 and RDT 3.0? Describe the FSM of RDT 3.0. (12)
c) Write short notes on (i) Persistent Timer and (ii) Keepalive Timer. (08)
d) What should be the relation between window sizes and sequence number in case of Selective Repeat? Why do we need the relationship? (05)
8. a) What are the shortcomings of symmetric key cryptography? How are those solved in public key cryptography. (06)
b) In symmetric key system, how do two entities establish shared secret key over a network? Explain a solution with suitable example. (10)
c) How does Multiple Access with Collision Avoidance (MACA) solve hidden terminal problem in wireless network? Explain with example. (07)
d) Suppose, A wants to send a packet to B. If A does not know B's MAC address then how does it determine this MAC address in the following two situations: (12)
i) A and B are in the same network.
ii) A and B are in the different network.

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
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Computer Networks

TIME: 1.5 hours

FULL MARKS: 120

- N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
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SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) What does protocol achieve? Differentiate between intranet and extranet. (10)
b) Why do we need Web caching? Discuss about the consistency of Web caching. (10)
c) What is P2P architecture. Describe what transport service does an app need? (10)
2. a) Compare among WWW, HTTP and Internet. Show the HTTP request and response message format. (10)
b) Suppose, *Alice* (ID – “alice@net.edu”) wants to send message to *Bob* (ID – “bob@denza.edu”). Show SMTP interactions between SMTP client and SMTP server for this communication. (20)
3. a) Consider the data bits $D = 11011011$ and Generator $G = 10110$. Calculate the Cyclic Redundancy Check bit R. (14)
b) How does RSA provide digital signature mechanism? (10)
c) Justify that – “Switch is more intelligent than hub”. (06)

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) Why do we need Network Address Translation (NAT)? Briefly explain how does NAT work. (10)
b) Compare the contrast link-state and distance vector routing algorithms. (09)
c) Suppose a source and a destination are connected through a switch and a router. If the source sends a message to the destination, then how do encapsulation and decapsulation happen in each layer of TCP/IP model of above-mentioned devices. Explain with necessary figure. (11)
5. a) Mention different components of Domain Name System (DNS). Explain various types of name server used in DNS by presenting their architecture and working mechanism. (11)
b) What is the relation between window size and sequence number in case of selective repeat? Why this relation is needed? (09)
c) List down the steps involved in SSL handshaking. Also present a pictorial view of packet transmission. (10)
6. a) Describe the technologies that are used for transitioning from IPv4 to IPv6. (06)
b) What is Ethernet? Describe how does CSMA/CD protocol work with a detail look on exponential back-off algorithm. (12)
c) Suppose, a company has 200 personal computers and 10 servers. The network administrator wants you to dynamically assign the IP address of those personal PCs. How personal computer get the IP address if you have a network address: 172.16.0.0/24. Also, state the dynamic range of IP addresses. (12)

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Computer Networks

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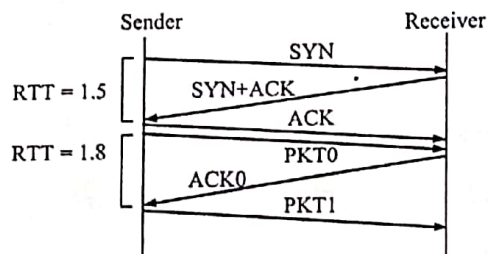
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SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is access network? Describe the enterprise access network with diagram. (08)
b) How does data pass through network in case of circuit switching? Explain in brief. (08)
c) What is throughput? What will be the throughput in case of Internet scenario? (06)
d) What is the differences between RDT 2.1 and RDT 2.2? Draw RDT 2.2 finite state machine for sender and receiver. Briefly describe this. (13)
2. a) Why is UDP used in transport layer though it is unreliable? (05)
b) In GB3, if every 5th packet is being transmitted is lost and if we have to send 10 packets, then how many transmissions are required? (11)
c) What is the relation between window size and sequence number in case of selective repeat? Why is this relation needed? (10)
d) Write short notes on the following terms: (09)
(i) Demultiplexing, (ii) Characteristics of UDP, and (iii) UDP segment format.
3. a) What is "syn-flooding" attack in case of connection establishment in TCP? Explain. (06)
b) How many ways are there to terminate a connection in TCP? Explain the ways in proper diagram. (12)
c) Suppose receiver application program consumes data as 1byte at a time. For this, the receiver TCP announces the window size as 1byte. So, the sending TCP sends 1byte at a time and the traffic intensity will be high. What has to be done to solve the problem? Explain two solutions. (06)
d) Explain Karn's algorithm in case of TCP timer. What will be the RTO of PKT1 in the following figure? (11)



4. a) How does TCP recognize the network congestion? Explain the Reno approach of managing this congestion by using necessary illustration. (13)
b) Explain block cipher and cipher-block chain algorithm in symmetric key cryptography. Why will we use cipher-block chain instead of block cipher? (14)
c) In case of RSA, if primes are $p = 5, q = 7$, public key is $(35, 5)$ and cipher text $c = 18$, then decrypt the cipher text to get plain text. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Suppose two nodes start to transmit at the same time a packet of length L over a broadcast channel of rate R . Denote the propagation delay between the two nodes as d_{prop} . What there be a collision if $d_{prop} > \frac{L}{R}$? Why or why not. (08)
b) Briefly describe that two dimensional parity checks can correct and detect a single bit error but a double bit error that can be detected but not corrected. (08)

- c) Why is an ARP query sent within a broadcast frame? Why is an ARP response sent within a frame with a specific MAC address? (07)
- d) You have to look carefully Figure 5(d). Computer A wants to send a frame to computer C'. (12)
Describe the two switches' behavior when A connects to C' and after completing the communication, how the switch table of both switches look like? (Initially switch table was empty). Finally, write down the self-learning algorithm of switch.

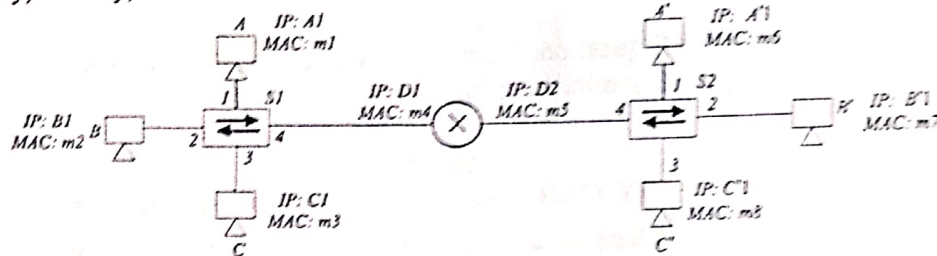
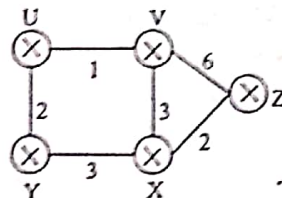


Figure 5(d)

6. a) Suppose you wanted to do a transaction securely from a remote client to a server as fast as possible. Would you use UDP or TCP? Why? (07)
- b) What is meant by handshaking protocol? Why is it said that FTP sends control information "out of band"? (07)
- c) What are the advantages and disadvantages of a circular DHT (with no shortcuts)? (08)
- d) Suppose the CSE department of KUET has 200 personal computer and 10 servers. The network administrator wants you to dynamically assign the IP address of those personal PC. How personal computer get the IP address if you have a network address 172.16.0.0/24. Also, state the dynamic range of IP addresses. (13)
7. a) Briefly describe different types of switching fabrics. Which, if any, can send multiple packets across the fabric in parallel? (06)
- b) Define HOL blocking and longest prefix matching with appropriate example. (08)
- c) Consider the network shown below and assume that each node initially knows the costs to each of its neighbors. Consider the distance vector algorithm and show the distance table entries at node Z. (12)



- d) Differentiate GET and conditional GET commands of HTTP. How does conditional GET command improve performance of web proxy and local cache mechanism? (09)
8. a) What is NAT traversal problem? How can you solve it? Explain briefly with appropriate figure. (11)
- b) Compare and contrast link-state and distance vector routing algorithms. (08)
- c) Describe the technologies that are used for transitioning from IPv6 to IPv4 to IPv6. (07)
- d) Consider the generator $G = 10011$ and suppose that D has the value 1010101010. What is the value of R? How can you detect the error using Cyclic Redundancy Check method? (09)

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CSE 4105
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SECTION A

(Answer ANY THREE questions from this section in Script A)

1. a) State the problems of a stop-n-wait type reliable data transfer protocol. Design a reliable, pipelined data transfer protocol that uses no buffer for out of order packets. (15)
b) Briefly describe the services provided by TCP. (09)
c) Show the format of HTTP request line. Mention at least two request type used by HTTP version 1.1 and explain. (06)
d) Explain common FTP commands. (05)
2. a) Consider transferring an enormous file of L bytes from host A to host B. What is the maximum value of L such that TCP sequence numbers are not exhausted? Assume an MSS of 1460 bytes. (05)
b) The TCP of a host receives five segments at the time shown (HH:MM:SS.mmm) below. If it has no data to send, show the acknowledgements sent by the host using line diagram. (09)
(i) at 0:0:0.000 (ii) at 0:0:0.027 (iii) at 0:0:0.400 (iv) at 0:0:1.200 (v) at 0:0:1.208
c) Show a sample SMTP interaction between SMTP client and server to transfer the message "Good morning my friend" from alice@xxx.com to bob@yyy.com. (11)
d) In TCP, what problems are solved by the following timers and how? (10)
(i) Persistence timer (ii) Time-waited timer
3. a) Suppose a DNS client is looking for the IP address of www.kuet.ac.bd. It already sent a query message to the DNS server. Show the response message sent by the server. Use necessary DNS compression. Assume the IP address is 128.175.13.63. (10)
b) Explain the necessity of mail access agent in email system. Compare and contrast the functionalities of POP and IMAP. (09)
c) Since UDP is unreliable, why is it used at transport layer? (05)
d) What do you mean by digital signature? How does RSA provides digital signature mechanism? (11)

4. a)

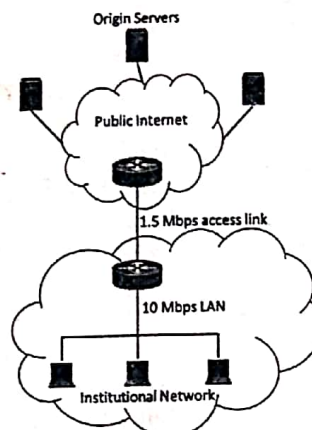


Figure 4(a)

Consider Figure 4(a), for which there is an institutional network connected to the internet. Let the average object size is 900 kbit and the average request rate from institutional browsers to the origin servers is 1.5 request per second. Also suppose that the amount of time it takes from when the router on the internet side of the access link forwards an HTTP request until it receives the response in two seconds on average. Model the total average response time as a sum of the average access delay and the average internet delay. For average access delay, use $\Delta/(1-\Delta\beta)$, where Δ is the time required to send an object over the access link and β is the arrival rate of objects to the access link.

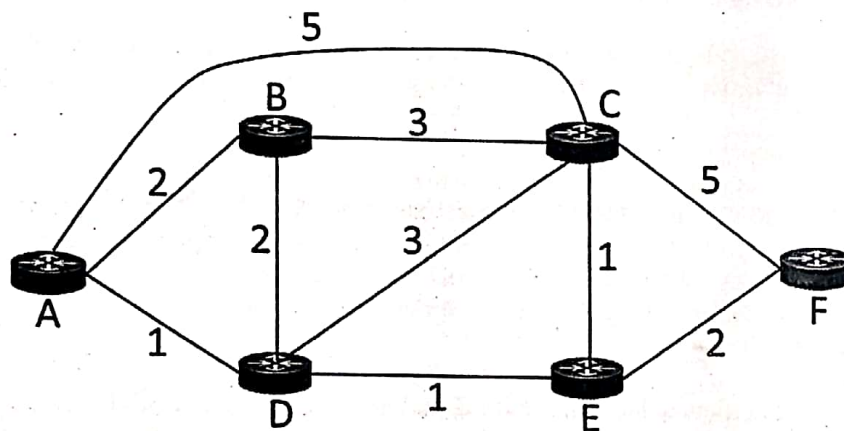
- i) Find the total average response time.
- ii) Now suppose a web cache is installed in the institutional LAN. Let the hit rate is 0.4. Find the total response time.

- b) In symmetric key system, how do two entities establish shared secret key over a network? Explain a solution with suitable example. (10)
- c) List down the steps involved in SSL handshaking. Also present a pictorial view of packets transmission. (10)

SECTION B

(Answer ANY THREE questions from this section in Script B)

5. a) How does header checksum can be calculated in IPv4 datagram? Why does it cover only data portion? Explain with example. (12)
- b) Define Computer Network. What are the major functionalities of Computer Network? (08)
- c) Apply Fragmentation on following information for IPv4 Datagram. (15)
- i) Original Datagram : 5000 bytes, Header : 20 bytes.
 - ii) MTU : First level 2000 bytes and Second level 800 bytes.
6. a) Concisely describe the error reporting messages of ICMP. (10)
- b) Determine the routing table/forwarding table at the router B from the following topology using a Link-State Routing Algorithm. (17)



- c) "Switch is more intelligent than hub" – justify the statement. (08)
7. a) A router inside an organization receives a packet with the destination 190.240.33.91. How does it find the subnetwork address to route the packet? Assume the subnet mask is /19. (05)
- b) Write a short note on Address Resolution Protocol (ARP). (09)
- c) What is count to infinity problem? Use a suitable example to show that the poisoned reverse technique can solve the problem. (12)
- d) Why do we need Network Address Translation (NAT)? Briefly explain how NAT works. (09)
8. a) Consider the data bits $D = 11011011$ and Generator $G = 10110$. Calculate the Cyclic Redundancy Check bit R . (12)
- b) How does Slotted ALOHA random access MAC protocol work? Show the efficiency of it. (13)
- c) Write down the services provided by the data link layer. (10)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
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CSE 4105

Computer Networks

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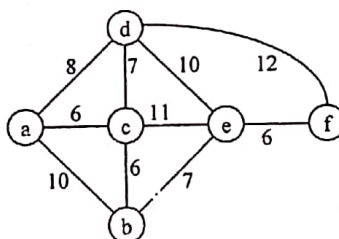
FULL MARKS: 210

- N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
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SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define statistical multiplexing. Which one is better between statistical multiplexing and Time Division Multiplexing? Explain with an appropriate example. (12)
b) Suppose source and destination are connected through a switch and a router. If source sends a message to destination then how do encapsulation and decapsulation happen in each layer of TCP/IP model of above mentioned devices? Explain with necessary figure. (11)
c) Suppose A is transmitting a packet to B. They are directly connected through a physical link. The length of the link is 1 km, the length of the packet is L bits and the transmission rate is 1 Mbps. (i) Now calculate how many bits will be arrived at B in 50 μ sec? (ii) If the packet length is 100 bytes then at what transmission rate R, the propagation delay would be equal to the transmission delay of packet? (12)
2. a) Consider sending a 2400 byte datagram into a link that has an MTU of 700 bytes. Suppose the original datagram is stamped with the identification number 422. How many fragments are generated? What are the values in the various fields in the IP datagram(s) generated related to fragmentation? (10)
b) Suppose the CSE department of KUET has 200 personal computers. The network administrator does not want to set the IP addresses statically. So, how the personal computers can get IP addresses? Explain steps of getting IP. (11)
c) Consider the following network. With the indicated link costs, use Dijkstra's shortest path algorithm to compute the shortest path from b to all network nodes. Show how the algorithm works by computing a table. (14)



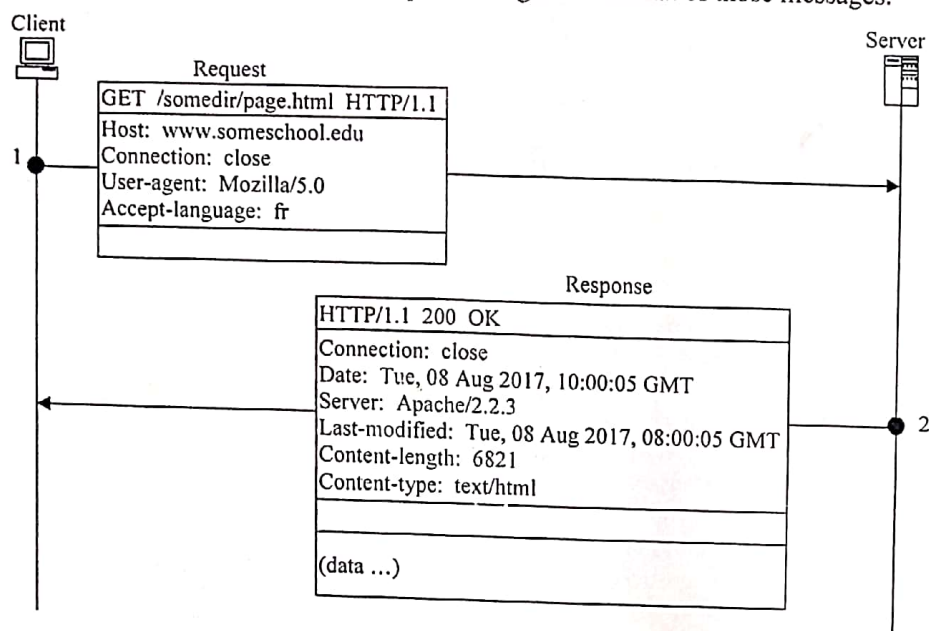
3. a) How does private IP addressing concepts resolve IPv4 address space problem? What are the new problems arises? Explain. (09)
b) Suppose A wants to send a packet to B. If A does not know B's MAC address then how does it determine this MAC address in the following two situations: (13)
i) A and B are in the same network.
ii) A and B are in the different network.
c) What is Ethernet? Describe the working principle of Ethernet's CSMA/CD protocol with a detail look on exponential back-off algorithm. (13)
4. a) Describe the technologies that are used for transitioning from IPv4 to IPv6. (06)
b) How does Multiple Access with Collision Avoidance (MACA) solve hidden terminal problem in wireless network? Explain with example. (08)
c) How does IEEE 802.11 deal with data fragmentation? Discuss. (10)
d) The CSE department of KUET just bought a Cisco switch which has 50 ports. The network administrator connected 40 personal computers to that switch. Now explain the learning mechanism of it. (11)

SECTION B

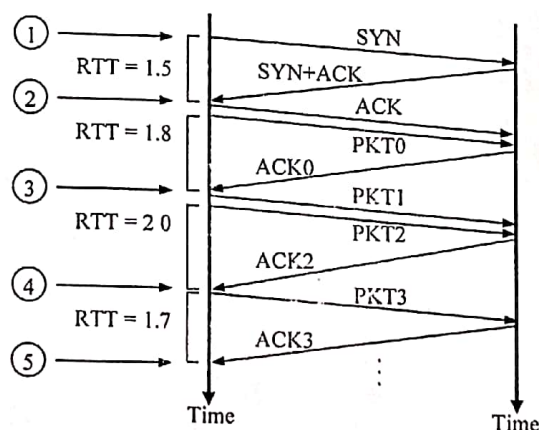
(Answer **ANY THREE** questions from this section in Script B)

5. a) Compare among the Internet, WWW and HTTP. (09)
b) What are the limitations of SMTP? How can these be solved by Multipurpose Internet Mail Extensions (MIME)? Briefly explain the working process and Base64 encoding of MIME. (11)

- c) Consider the following communication between a client and a server. The figure shows the different messages transferred between them. Explain each section as well as the statements that are presented in the messages. Also provide a general format of those messages. (15)



6. a) Mention different components of Domain Name System (DNS). Explain various types of name server used in DNS by presenting their architecture and working mechanism. (11)
- b) Design a stop-n-wait type reliable data transfer protocol by showing a finite state machine. Also present the line diagrams to explain its operation for various packet transmission scenarios. (14)
- c) How does selective repeat protocol work? Explain. (10)
7. a) How does TCP recognize the network congestion? Explain, with necessary illustration, the Reno approach of managing this congestion. (14)
- b) What is silly window syndrome? How does TCP handle this issue? (08)
- c) How does TCP set the timeout interval for retransmission timer? What are the Jacobson/Karels proposals for setting that interval? Consider the following communication scenario of TCP. Figure shows the sample RTTs in seconds. Compute the timeout intervals at the points shown by arrows i.e. at initial position and at the end of each acknowledge received. (13)



8. a) Design a protocol to authenticate a live entity. Can it be used for public key system? Why or why not? (12)
- b) Let you are a network administrator, design a rule set for your packet filter firewall to allow your local users only to browse outside and no other traffic except the locally initiated return traffic from outside is allowed to enter your network. (07)
- c) Briefly explain how Secure Socket Layer (SSL) works on top of TCP and what services it provides. (09)
- d) What are the basic differences between TCP and UDP? (07)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
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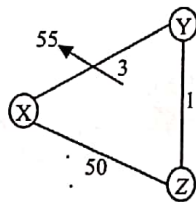
FULL MARKS: 210

- N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
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SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is layering in computer networks? Write down the major functions of network layer. (07)
b) What are the advantages of packet switching over circuit switching? (06)
c) Suppose a TCP message that contains 2048 bytes of data and 20 bytes of TCP header is passed to IP for delivery across two networks of the Internet (i.e., from the source host to a router to the destination host). The first network uses 14-bytes headers and has an MTU of 1024 bytes; the second user 8-byte header with an MTU of 512 bytes. Give the sizes and offsets of the sequence of fragments delivered to the network layer at the destination host. Assume all IP header are 20 bytes. (12)
d) Differentiate between IPv4 and IPv6 header format. Explain the technologies that are used for transition from IPv4 to IPv6. (10)
2. a) Write down the pros and cons of static and dynamic routing. Explain link-state routing algorithm with a proper example. (13)
b) What is DHCP? How does DHCP work? Write down the advantages and disadvantages of DHCP. (12)
c) What does ICMP stand for? What does it do? Describe the error reporting messages of ICMP. (10)
3. a) "A transport-layer protocol provides logical communication between processes running on different hosts, a network-layer protocol provides logical communication between hosts."- Justify the statement. (08)
b) What are the essential tools of RDT protocol? Describe the working mechanism of RDT 2.0. How can the performance problem of RDT be solved? (13)
c) Describe the TCP timers in Brief. (08)
d) Since UDP is unreliable, why is it used at transport layer? (06)
4. a) Consider the following graph. Assume that link (x,y) cost has been changed from 3 to 55. (11)
Show that the poisoned reverse technique can resolve the count to infinity problem in this case.



- b) In theory TCP is fair, but in face it is not. Give an example. (07)
- c) What is the NAT traversal problem for P2P applications, say Skype? (05)
- d) Suppose the values of p and q are 7 and 13 respectively. Find suitable public and private keys for RSA algorithm. Verify the keys for the message "CSE". (12)

SECTION B

(Answer ANY THREE questions from this section in Script B)

5. a) What are the techniques to resolve IPv4 address space problem? Explain in Brief. (10)
b) What is digital signature? How authentication is provided using digital signature. (10)
c) What is Diffie Hellman key exchange algorithm? Explain its working principle with an example. (10)
d) What do firewalls do? Explain shortly. (05)
6. a) "Sockets are the software interface"-Justify the statement. (05)
b) What are the transport services does an application need? Briefly describe them. (10)
c) What is the benefit of stateless property of HTTP server? How HTTP servers identify users with stateless property? Explain briefly. (10)
d) Differentiate GET and Conditional GET commands of HTTP. How does conditional GET command improve performance of web proxy and local cash mechanism? (10)
7. a) Why is it said that FTP sends control information "out of band"? (05)
b) Suppose Alice, with a web-based email account (such as gmail) sends a message to Bob, who accesses his mail from his mail server using POP3. Discuss how the message gets from Alice's host to Bob's host. Be sure to list the series of application layer protocols that are used to move the message between the two hosts. (10)
c) Write filtering/forwarding algorithm of a switch. List the actions with corresponding condition(s) that a switch may perform when it receives a frame. (08)
d) What is service provided by DNS system? Explain load distribution briefly. (06)
e) If all the links in the Internet were to provide reliable delivery service, would the TCP reliable delivery service be redundant? Why or why not? (06)
8. a) Consider an e-commerce site that wants to keep a purchase record for each of its customer. Describe how this can be done with cookies? (08)
b) Why are content distribution networks (CDNs) important in internet system? Describe the role of DNS in CDN. (07)
c) Write down the advantages and disadvantages of polling and Token passing MAC protocol. (09)
d) Discuss Byte stuffing with an appropriate example. (06)
e) Why a network interface card is masked as semi-autonomous device? (05)

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SECTION A

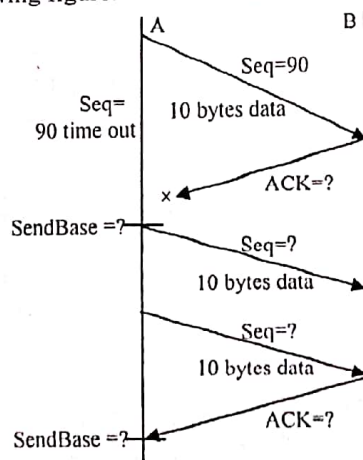
(Answer **ANY THREE** questions from this section in Script A)

1. a) Define computer network and Internet. Explain Internet on service view points. (09)
- b) Why is it said that packet switching employs statistical multiplexing? Contrast statistical multiplexing with the multiplexing that takes place in TDM and FDM. (09)
- c) "Message segmentation reduces the end to end delay increasing resource utilisabilities"- Justify the statement. (08)
- d) Explain Internet structure with different tiers. What is the benefit of tierwise ISP connections for internet service? (09)
2. a) Why does TCP socket identification need 4-tuple instead of 2-tuple? What are those tuples? (08)
- b) Why UDP is called a bare-bone protocol in compare of TCP? (06)
- c) In respect of Rdt 2.0, briefly state the objective of the following used techniques: (i) Checksum (ii) ACK packet (iii) NAK packet. (09)
- d) Consider sending a file of 10KB packet with 1 Gbps bandwidth link and 15 msec end-to-end propagation delay. Answer the following questions: (12)
 - (i) What is the required transmission time?
 - (ii) What is the channel utilization (U)?

$$\text{Note that } U = \frac{\frac{L}{R}}{RTT + \frac{L}{R}}$$

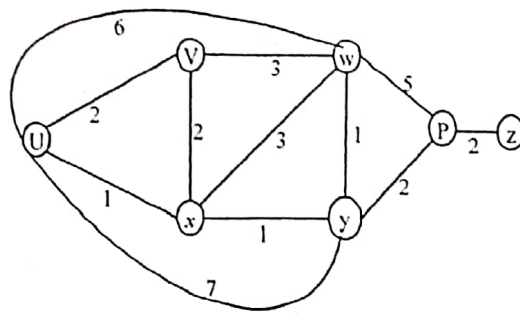
(iii) For getting the channel utilization 0.5 what will be the transmission time?

3. a) What is the performance problem of Rdt 3.0 (stop-and-wait) protocol? How this problem can be resolved? (07)
- b) Sketch a diagram where selective repeat protocol experiences a dilemma. Give your opinion how this dilemma can be overcome. (10)
- c) Consider TCP's communication scenario between A and B. A and B have initial sequence number 90 and 180 respectively. A is sending a file to B with a packet size of 10 bytes in the following figure. Determine the values of seq#, ACK# and sendBase#. (08)

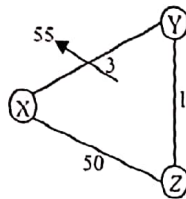


- d) What do you mean by NAT? Explain how NAT operation is performed. (10)

4. a) What is the HOL blocking in the switch fabric of a router? (05)
 b) Consider the following graph. Using Link-state algorithm find the shortest-path tree for router U. (12)



- c) Consider the following graph. Assume that link (x,y) cost has been changed from 3 to 55. Show that the poisoned reverse technique can resolve the count to infinity problem in this case. (11)



- d) Is RIP Intra-AS or Inter-AS routing protocol? Show a simple example of RIP routing table entries. (07)

SECTION B

(Answer ANY THREE questions from this section in Script B)

5. a) Briefly explain Nonpersistent HTTP and Persistent HTTP with/without pipeline. Compare TCP connections among the three HTTP methods for a HTML file containing 10 other objects' link. (10)
 b) Differentiate GET and conditional GET commands of HTTP. How does conditional GET command improve performance of web proxy and local cache mechanism? (08)
 c) Does web browsing require UDP service along with TCP? Explain briefly. (05)
 d) "Email demands both push and pull operations"-Justify the statement. (06)
 e) Compare SMTP, HTTP and DNS on the basis of direct transfer and transfer through intermediate hosts. (06)
6. a) Explain common FTP commands and replies. (07)
 b) Explain importance of DNS in service view points. Also explain iterative and recursive DNS queries with appropriate example. (10)
 c) How DNS and CDN together and alone provide better web browsing facilities? (08)
 d) Explain importance of Network Operations Communications Center (NOCC) in content Distribution Network. (05)
 e) "In computer networking switch is transparent to hosts but router is not"-is it true? Justify your answer. (05)
7. a) Why a network interface card is a marked as semi autonomous device? (05)
 b) Explain ideal Multiple Access (MAC) protocol scenario. Compare three different MAC protocols. (09)
 c) Distinguish slotted ALOHA and pure ALOHA protocols. (06)
 d) Write filtering/forwarding algorithm of a switch. List the actions with corresponding condition(s) that a switch may perform when it receives a frame. (09)
 e) Compare routing switches and hubs on operational viewpoints. (06)
8. a) Why does a computer need an MAC address, inspite of having an IP address? (05)
 b) "MAC address of a host never goes outside the network"-is it true? Justify your answer. (05)
 c) Briefly explain various desirable properties of secure communication. (06)
 d) What are shortcomings of symmetric key cryptography? How are those solved in public key cryptography? (07)
 e) What do you mean by digital signature? How does RSA provides digital signature mechanism? (12)